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INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Information Technology

End-Term Examination Nov – 2025

Course : B.Tech

Semester: 3rd

Subject Code: IT 3002

Subject Name: DBMS

Branch: IT

Max. Marks:60

Time: 03hrs

NOTE:

1. All the questions are compulsory.
2. While solving write question number and its part clearly.
3. Solve questions in given sequence and all the sub parts should be at same place.

SN	Question	MM
1	(A) Consider the following educational institute database and express the following queries in relational algebra: (Note: Primary key are bold) Student (S_Id, Name, Dept_Name, Total_Credit) Department (Dept_Name, Building, Budget) Course (Course_Id, Title, Dept_Name, Credits) Prereq (Course_Id, Prereq_Id) Instructor (I_Id, Name, Dept_Name, Salary) Advisor (S_Id, I_Id) Time_Slot (Time_Slot_Id, Day, Start_Time, End_Time) Classroom (Building, Room_Number, Capacity) Section (Course_Id, Sec_Id, Semester, Year, Building, Room_Number, Time_Slot_Id) Teaches (I_Id, Course_Id, Sec_Id, Semester, Year) Takes (S_Id, Course_Id, Sec_Id, Semester, Year, Grade) 1. Find the names of departments that offer no courses requiring a prerequisite. 2. Find the names of instructors who earn a salary greater than at least one other instructor in the same department. 3. Find the names of students who have taken all courses offered by the 'Comp. Sci.' department. 4. Find the names of all instructors who have never taught a course. 5. Find the names of students who have enrolled only for 'IT-101' and 'CSE-200'. (B) Consider a relation R(P, Q, R, S, T), with following set of functional dependencies F = {PQ → RS, P → R, R → T, PS → Q}, Find the canonical cover.	8+4
2	(A) Consider the following Airline management database and express the following queries in SQL: (Note: Primary key are underline) Passengers(<u>PassengerID</u> , FirstName, LastName, DOB, Gender, Email, Phone, PassportNo) Airports(<u>AirportCode</u> , Name, City, Country, Timezone) Aircrafts(<u>AircraftID</u> , Model, Manufacturer, TotalSeats, Range_km) Routes(<u>RouteID</u> , OriginAirportCode, DestAirportCode, Distance_km) Flights(<u>FlightID</u> , FlightNumber, RouteID, AircraftID, ScheduledDeparture, ScheduledArrival, Status) Bookings(<u>BookingID</u> , PassengerID, BookingDate, TotalAmount, BookingStatus) Tickets(<u>TicketID</u> , BookingID, FlightID, SeatNumber, FareClass, FareAmount, TicketStatus) CrewMembers(<u>CrewID</u> , FirstName, LastName, Role, LicenseNo) FlightCrew(<u>FlightID</u> , CrewID , AssignmentRole) 1. Find passengers whose last name starts with "Singh" (pattern matching): 2. Top 5 upcoming flights with most booked seats. 3. List tickets for a particular passenger sorted by departure time: 4. Find routes with average load (booked seats / aircraft seats) > 70% (requires join to Aircrafts): 5. List tickets for a particular passenger sorted by departure time. (B) Consider a relation R(A, B, C, D, E, F) with following set of functional dependencies holding on it: {AB → C, AD → E, B → D, E → A}. Find the highest normal form of relation R. If the relation is not in BCNF then decompose it in BCNF relations and check whether the decomposition is dependency preserving or not.	8+4
3	(A) Consider the following Airline management database and express the following queries in Tuple calculus: (Note: Primary key are bold) Employee(EmpID, FirstName, LastName, DOB, Gender, Email, Phone, HireDate, Salary, DeptID, ManagerID) Department(DeptID, DeptName, Location, HeadEmpID) Project(ProjID, ProjName, StartDate, EndDate, Budget, DeptID)	4+8

WorksOn(EmpID, ProjID, HoursPerWeek)

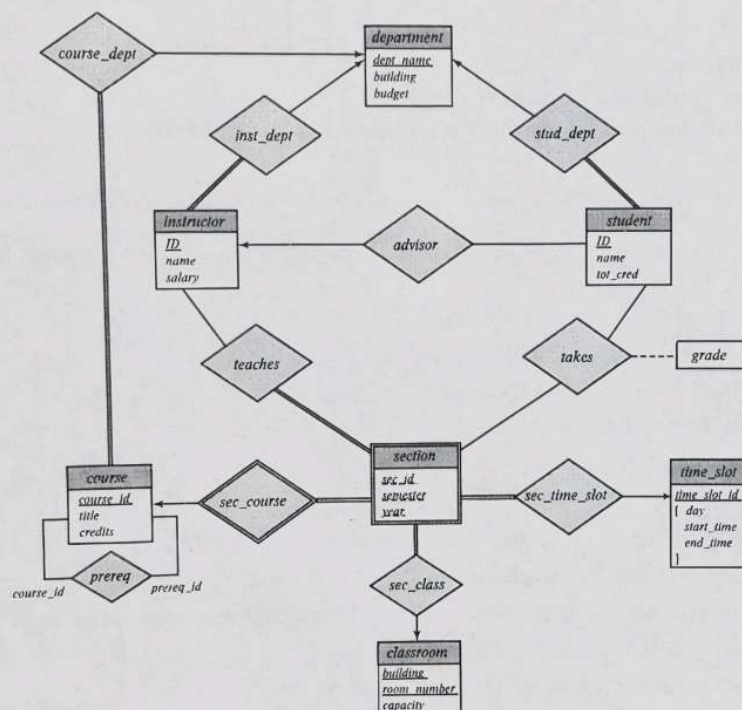
Attendance(AttID, EmpID, Date, CheckInTime, CheckOutTime, Status)

SalaryHistory(SalHistID, EmpID, FromDate, ToDate, Salary)

LeaveRequest(LeaveID, EmpID, FromDate, ToDate, LeaveType, Status)

1. Employees who work in a department located in "Delhi".
2. Employees who work on every project of their department.
3. Employees who have never worked on any project outside their own department.

(B) Using the ER diagram described below, convert the design into a fully normalized relational schema. Identify all primary keys, foreign keys, and relationship tables clearly.



4

(A) Consider university database relation R(StudentID, CourseID, InstructorID, RoomNo, TimeSlot, DeptID, Credits) to store course allocation information, with following functional dependencies:

- StudentID, CourseID → InstructorID, RoomNo, TimeSlot
- InstructorID → DeptID
- CourseID → Credits
- DeptID, RoomNo → TimeSlot

Because the relation contains multiple attributes with overlapping dependencies, the database administrator proposes decomposing R into the following sub-relations:

- R1(StudentID, CourseID, InstructorID)
- R2(InstructorID, DeptID)
- R3(CourseID, Credits)
- R4(RoomNo, DeptID, TimeSlot)

Using the given relation, functional dependencies, and the proposed decomposition, evaluate whether the decomposition $R \rightarrow \{R1, R2, R3, R4\}$ is both lossless-join and dependency-preserving.

(B) Consider the following description of an International Music Association, an organization that maintains information about its member orchestras, their musicians, and the recordings they produce. For each orchestra, the system stores its name, city, country, and music director. For each musician, the system records a unique musician number, name, the instruments they play, and annual salary. A musician belongs to exactly one orchestra, while an orchestra may consist of many musicians. Musicians may also hold a music degree, and for each degree the system stores the degree name, university, and year. Each musician may possess only a single degree. Each music composition is created by a composer. For each composer, the system records the composer name (which uniquely identifies the composer), country of origin, and date of birth. A composer may create multiple compositions; however, composition names are unique only within the same composer. For each composition, the system also stores the year of composition. The system must also track which orchestras have recorded which compositions, and which compositions have been recorded by which orchestras. For each recording, the year of recording and the price must be stored. Based on this description, draw an ER diagram for the organization and clearly mention any assumptions you make.

4+8

5 (A) A distributed inventory management system allows multiple warehouse servers to process purchase orders, update stock levels, and generate shipment records concurrently. To maintain database consistency across distributed nodes, the system uses the Two-Phase Locking (2PL) protocol for concurrency control. Consider the following scenario: - Transaction T1 updates the stock quantity of an item, reads supplier information, and then inserts a shipment request. - Transaction T2 simultaneously attempts to read the same stock quantity, check back-order status, and update customer demand records. - Transaction T3 attempts to update the same item's supplier information after T1 has already locked it. (a) Analyze how basic 2PL determines the order in which locks are granted to T1, T2, and T3. Clearly explain the growing and shrinking phases for each transaction in this scenario. (b) Identify and discuss at least two possible concurrency anomalies that could occur if 2PL were not used, and explain why 2PL prevents them. (c) Demonstrate how a deadlock may arise among T1, T2, and T3 under basic 2PL, showing the exact lock acquisition sequence leading to the cycle. (d) Recommend and justify an appropriate deadlock-handling strategy (e.g., wait-die, wound-wait, or deadlock detection with a wait-for graph) for this environment. (e) Evaluate how the use of Strict 2PL or Rigorous 2PL would change the outcome of this scenario, particularly with respect to cascading rollbacks and commit ordering.	8+4
(B) Consider the three transactions T1 T2 and T3 and the schedules given below. Check whether the given schedules are serializable or not. If schedule is serializable, write down equivalent serial schedule(s). $S_1: r_1(X); r_2(Z); r_1(Z); r_3(X); r_3(Y); w_1(X); \text{commit}_1; w_3(Y); \text{commit}_3; r_2(Y); w_2(Z); w_2(Y); \text{commit}_2;$ $S_2: r_1(X); r_2(Z); r_1(Z); r_3(X); r_3(Y); w_1(X); w_3(Y); r_2(Y); w_2(Z); w_2(Y); \text{commit}_1; \text{commit}_2; \text{commit}_3;$	

*****Best of Luck*****